

CENTRAL BOARD OF SECONDARY EDUCATION

HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)

SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 7: WORK, ENERGY, AND SIMPLE MACHINES [SET B - ADVANCED HOTS]

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25

PAPER CODE: 086/CH07/SET-B

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET B - HIGHER ORDER THINKING SKILLS):

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Advanced Conceptual and Numerical MCQs with single correct option.
3. **Section B (Q16 – Q20):** Critical Assertion-Reasoning based questions testing analytical clarity.
4. **Section C (Q21 – Q25):** Case-Study / Practical Experimental Data-interpretation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: ADVANCED & NUMERICAL MCQS (Q.1 TO Q.15)

[15 MARKS]

Q.1 An electric water pump with an efficiency of 80% lifts 1200 kg of water to a height of 20 m in 2 minutes ($g = 10 \text{ m/s}^2$). What is the electrical power input to the pump?

- (A) 2000 W (B) 2500 W
(C) 1600 W (D) 3200 W

Q.2 The kinetic energy of an object is increased by 300% (becoming 4 times its original value). By what percentage does its linear momentum increase?

- (A) 100% (B) 200%
(C) 300% (D) 50%

Q.3 An ideal block and tackle pulley system consists of 5 pulleys (velocity ratio $VR = 5$). What effort force is needed to lift a load of 1000 N?

- (A) 5000 N (B) 200 N
(C) 250 N (D) 500 N

Q.4 If the efficiency of the above 5-pulley system is actually 80% due to friction and cable weight, what effort is required?

- (A) 200 N (B) 250 N
(C) 300 N (D) 160 N

Q.5 A spring with force constant $k = 400 \text{ N/m}$ is compressed by 10 cm (0.10 m). What is the elastic potential energy stored in the spring ($E_p = \frac{1}{2} kx^2$)?

- (A) 40 J (B) 2 J
(C) 4 J (D) 20 J

Q.6 A horizontal force F pushes a 4 kg box along a straight path where force varies with displacement: $F = 10 \text{ N}$ from $s=0 \text{ m}$ to $s=4 \text{ m}$, and then decreases linearly to 0 at $s=8 \text{ m}$. What is the total work done?

- (A) 40 J (B) 80 J
(C) 60 J (D) 50 J

Q.7 A simple pendulum bob of mass 0.2 kg is released from rest at a height of 0.8 m above its lowest equilibrium position. What is its speed at the lowest point ($g = 10 \text{ m/s}^2$)?

- (A) 2 m/s (B) 4 m/s
(C) 8 m/s (D) 16 m/s

Q.8 A crowbar of length 150 cm is used to displace a heavy rock. The fulcrum is placed 30 cm from the rock (load). What is the ideal mechanical advantage of this Class 1 lever?

- (A) 5 (B) 4
(C) 3 (D) 0.2

Q.9 A 50 kg girl climbs up a flight of 40 stairs, each 15 cm high, in 20 seconds ($g = 10 \text{ m/s}^2$). Her average power output is:

- (A) 150 W (B) 300 W
(C) 1500 W (D) 3000 W

Q.10 A car travelling at 20 m/s skids to a halt over 40 m when brakes lock. If the same car travels at 40 m/s (twice the speed), what will be the skidding distance under identical tire-road friction?

- (A) 80 m (B) 120 m
(C) 160 m (D) 200 m

Q.11 Which of the following forces does negative work on an object in motion?

- (A) Gravitational force on an apple falling from a tree (B) Friction acting on a skidding bicycle tire
(C) Centripetal force on a satellite orbiting Earth (D) Tension in string pulling a toy cart forward

Q.12 An inclined plane has length $L = 5 \text{ m}$ and height $h = 1 \text{ m}$. Neglecting friction, what is its ideal mechanical advantage?

- (A) 5 (B) 0.2
(C) 4 (D) 1

Q.13 A machine with an efficiency of 75% delivers 1500 J of useful work. What was the total work energy input to the machine?

- (A) 1125 J (B) 2000 J
(C) 2500 J (D) 1800 J

Q.14 An object of mass m is moving at speed v . Another object has mass $2m$ and speed $v/2$. What is the ratio of their kinetic energies?

- (A) 1 : 1 (B) 2 : 1
(C) 1 : 2 (D) 4 : 1

Q.15 A consumer uses five 100-Watt incandescent light bulbs for 6 hours daily for 30 days. How many commercial units (kWh) of electricity are consumed?

- (A) 90 units (B) 180 units
(C) 900 units (D) 45 units

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).

- (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): No machine can ever have an efficiency exceeding 100%.

Reason (R): Energy cannot be created according to the universal law of conservation of energy, and some input energy is inevitably lost as heat due to friction.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): The work done by a centripetal force on an object in uniform circular motion is always zero.

Reason (R): The centripetal force acts towards the center of the circle, perpendicular to the instantaneous tangential displacement vector.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): Two persons of masses 50 kg and 80 kg who climb the same 10-meter staircase do different amounts of work.

Reason (R): Work against gravity is directly proportional to mass ($W = mgh$).

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): When a fast-moving car hits a spring barrier, the kinetic energy of the car is converted into elastic potential energy of the compressed spring.

Reason (R): Total mechanical energy is conserved in the absence of dissipative friction.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): An astronaut inside an orbiting space station feels weightless because gravitational force from Earth is zero in orbit.

Reason (R): Gravity drops to zero as soon as an object enters space beyond Earth's atmosphere.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) Both (A) and (R) are false.

SECTION C: EXPERIMENTAL DATA & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Energy Auditing of an Electric Locomotive

A freight railway locomotive engine has a rated power output of 3.0 Megawatts (3×10^6 W). It hauls a fully loaded train of total mass 2×10^6 kg (2000 tonnes) up a gentle mountain gradient of 1 in 100 (height increases by 1 m for every 100 m of track). The train maintains a constant speed of 20 m/s (72 km/h). Take $g = 10 \text{ m/s}^2$.

Q.21 In one second, the train covers a track distance of 20 m. What vertical height does the train gain in one second?

- (A) 0.1 m
 (B) 0.2 m
 (C) 0.5 m
 (D) 1.0 m

Q.22 What is the rate of work done against gravity (power needed to lift the train mass) per second?

(A) $4.0 \times 10^6 \text{ W}$

(B) $4.0 \times 10^5 \text{ W}$

(C) $2.0 \times 10^6 \text{ W}$

(D) $1.0 \times 10^6 \text{ W}$

Q.23 If the locomotive provides a traction force of $150,000 \text{ N}$ to overcome rolling friction and air drag at 20 m/s , what power is expended against friction?

(A) $3.0 \times 10^6 \text{ W}$ (3 MW)

(B) $1.5 \times 10^6 \text{ W}$

(C) $6.0 \times 10^6 \text{ W}$

(D) $750,000 \text{ W}$

Q.24 How much kinetic energy does the 2000-tonne train possess while cruising at 20 m/s ?

(A) $4.0 \times 10^8 \text{ J}$ (400 MJ)

(B) $2.0 \times 10^8 \text{ J}$

(C) $8.0 \times 10^8 \text{ J}$

(D) $4.0 \times 10^7 \text{ J}$

Q.25 When braking on downhill slopes, modern trains utilize regenerative brakes that convert kinetic energy into electrical energy returned to the power grid. This confirms which universal law?

(A) Law of conservation of mechanical momentum

(B) Law of conservation of energy

(C) Newton's second law

(D) Coulomb's law

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET B)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & IN-DEPTH EXPLANATIONS (SET B)

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.1	(B)	$SW = mgh = 1200 \times 10 \times 20 = 240,000 \text{ J}$. Time $t = 120 \text{ s}$. $SP_{\text{output}} = \frac{240,000}{120} = 2000 \text{ W}$. $SP_{\text{input}} = \frac{P_{\text{output}}}{0.80} = \frac{2000}{0.80} = 2500 \text{ W}$.
Q.2	(A)	$p = \sqrt{2m E_k}$. If E_k quadruples ($4 E_k$), p becomes $\sqrt{4} = 2p$ (doubles). An increase from p to $2p$ is a 100% increase.
Q.3	(B)	For an ideal system, $MA = VR = 5$. $\text{Effort} = \frac{\text{Load}}{MA} = \frac{1000}{5} = 200 \text{ N}$.
Q.4	(B)	$MA = \eta \times VR = 0.80 \times 5 = 4$. $\text{Effort} = \frac{\text{Load}}{MA} = \frac{1000}{4} = 250 \text{ N}$.
Q.5	(B)	$E_p = \frac{1}{2} k x^2 = \frac{1}{2} \times 400 \times (0.10)^2 = 200 \times 0.01 = 2 \text{ Joules}$.
Q.6	(C)	Area under F - s graph: Rectangle from 0 to 4 m $= 10 \times 4 = 40 \text{ J}$. Triangle from 4 to 8 m $= \frac{1}{2} \times (8 - 4) \times 10 = 20 \text{ J}$. Total work $= 40 + 20 = 60 \text{ J}$.
Q.7	(B)	Conservation of energy: $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \times 10 \times 0.8} = \sqrt{16} = 4 \text{ m/s}$.
Q.8	(B)	$\text{Load arm} = 30 \text{ cm}$. $\text{Effort arm} = 150 - 30 = 120 \text{ cm}$. $MA = \frac{\text{Effort arm}}{\text{Load arm}} = \frac{120}{30} = 4$.
Q.9	(A)	$h = 40 \times 0.15 = 6 \text{ m}$. Work $W = mgh = 50 \times 10 \times 6 = 3000 \text{ J}$. Power $P = \frac{3000}{20} = 150 \text{ W}$.
Q.10	(C)	Work done by friction $= \text{Initial } E_k \Rightarrow f \cdot s = \frac{1}{2}mv^2 \Rightarrow s \propto v^2$. Doubling velocity quadruples stopping distance: $40 \times 4 = 160 \text{ m}$.
Q.11	(B)	Friction opposes displacement ($\theta = 180^\circ$). $W = F s \cos 180^\circ = -Fs$, doing negative work that converts kinetic energy into thermal energy.
Q.12	(A)	For an inclined plane, $MA = \frac{\text{Length}}{\text{Height}} = \frac{5}{1} = 5$.
Q.13	(B)	$\text{Input} = \frac{\text{Output}}{\eta} = \frac{1500}{0.75} = 2000 \text{ J}$.
Q.14	(B)	$E_1 = \frac{1}{2}mv^2$. $E_2 = \frac{1}{2}m(v/2)^2 = m(v^2/4) = \frac{1}{4}mv^2 = \frac{1}{2}E_1$. Hence, $E_1 : E_2 = 2 : 1$.
Q.15	(A)	Total power $= 5 \times 100 = 500 \text{ W} = 0.5 \text{ kW}$. Energy $= 0.5 \text{ kW} \times 6 \text{ h/day} \times 30 \text{ days} = 90 \text{ kWh (units)}$.
Q.16	(A)	Because energy output $\leq$ energy input, efficiency $\eta = \frac{W_{\text{out}}}{W_{\text{in}}} \times 100 \leq 100\%$. Friction and heat dissipation make $\eta < 100\%$ in all real machines.
Q.17	(A)	$W = F s \cos \theta$. Since centripetal force is orthogonal to tangential displacement ($\theta = 90^\circ$), $\cos 90^\circ = 0$, so no work is done.
Q.18	(A)	The 80 kg person lifts greater weight through the same height, doing $80 \times 10 \times 10 = 8000 \text{ J}$, while the 50 kg person does 5000 J .
Q.19	(A)	The conservative restoring force of the spring decelerates the car, converting initial kinetic energy $\frac{1}{2}mv^2$ into stored spring energy $\frac{1}{2}kx^2$.
Q.20	(D)	Both statements are completely false! Earth's gravity at 400 km altitude is still $\approx 90\%$ of surface gravity. The weightlessness feeling arises because the station and astronaut are in perpetual free fall around Earth.
Q.21	(B)	$\text{Vertical height per second} = 20 \text{ m} \times \frac{1}{100} = 0.2 \text{ m}$.
Q.22	(A)	$SP_{\text{gravity}} = \frac{mgh}{t} = 2 \times 10^6 \times 10 \times 0.2 = 4.0 \times 10^6 \text{ W (4.0 MW)}$... wait, if gradient is 1 in 100, $h = 0.2 \text{ m}$, $m = 2 \times 10^6 \text{ kg}$, $mgh = 2 \times 10^6 \times 10 \times 0.2 = 4 \times 10^6 \text{ W}$.
Q.23	(A)	$P = F \times v = 150,000 \text{ N} \times 20 \text{ m/s} = 3,000,000 \text{ W} = 3.0 \text{ MW}$.

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.24	(A)	$E_k = \frac{1}{2} m v^2 = \frac{1}{2} (2 \times 10^6) (20^2) = 10^6 \times 400 = 400 \times 10^6 \text{ J} = 400 \text{ MJ}.$
Q.25	(B)	Energy cannot be destroyed; regenerative braking captures kinetic energy and converts it into reusable electrical energy via generators.